

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 1

Name	Muhammad Eusoff Aminurrashid Bin Shukri
Student ID	1211112228
Section	SAM1, 1A

1. You need to install Linux server in the Lab PC using VirtualBox. Identify the **minimum** requirement on RAM and disk space for THREE (3) most popular Linux server namely Fedora, OpenSuSE and Ubuntu. Compare the requirements of OpenBSD, FreeBSD and NetBSD.

Distro	Fedora	OpenSuSE	Ubuntu	OpenBSD	FreeBSD	NetBSD
RAM	2GB	2GB	4GB	256MB	512MB	64MB
HDD	8.95GB	7.59GB	6.06GB	670.05MB	1.09GB	622.42MB

2. Identify the CPU type and speed, RAM size and hard disk size of your PC/notebook. Use and show command/program in Windows or Linux to be used during the identification process.

Device name AMENGSZ

Processor 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz

Installed RAM 20.0 GB (19.7 GB usable)

The screenshot shows the Windows Disk Management utility. At the top, system information is displayed: Device name AMENGSZ, Processor 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz, and Installed RAM 20.0 GB (19.7 GB usable). Below this, the Disk Management window shows a list of volumes. The volumes are: (Disk 0 partition 1) 260 MB, (Disk 0 partition 5) 1000 MB, (Disk 0 partition 6) 260 MB, Data (D:) 64.00 GB, and OS (C:) 411.44 GB. The bottom section shows a detailed view of Disk 0, which is a 476.92 GB Basic disk. It contains several partitions: a 260 MB Healthy (EFI) partition, a 411.44 GB NTFS (BitLocker Encrypted) partition (OS (C:)), a 64.00 GB NTFS (BitLocker Encrypted) partition (Data (D:)), a 1000 MB Healthy (Recovery) partition, and a 260 MB Healthy (Recovery) partition. A legend at the bottom indicates that black represents Unallocated space and blue represents Primary partitions.

Volume	Layout	Type	File System	Status	Capacity	Free Space	% Free
(Disk 0 partition 1)	Simple	Basic		Healthy (E...)	260 MB	260 MB	100 %
(Disk 0 partition 5)	Simple	Basic		Healthy (R...)	1000 MB	1000 MB	100 %
(Disk 0 partition 6)	Simple	Basic		Healthy (R...)	260 MB	260 MB	100 %
Data (D:)	Simple	Basic	NTFS (BitLo...)	Healthy (B...)	64.00 GB	19.91 GB	31 %
OS (C:)	Simple	Basic	NTFS (BitLo...)	Healthy (B...)	411.44 GB	108.12 GB	26 %

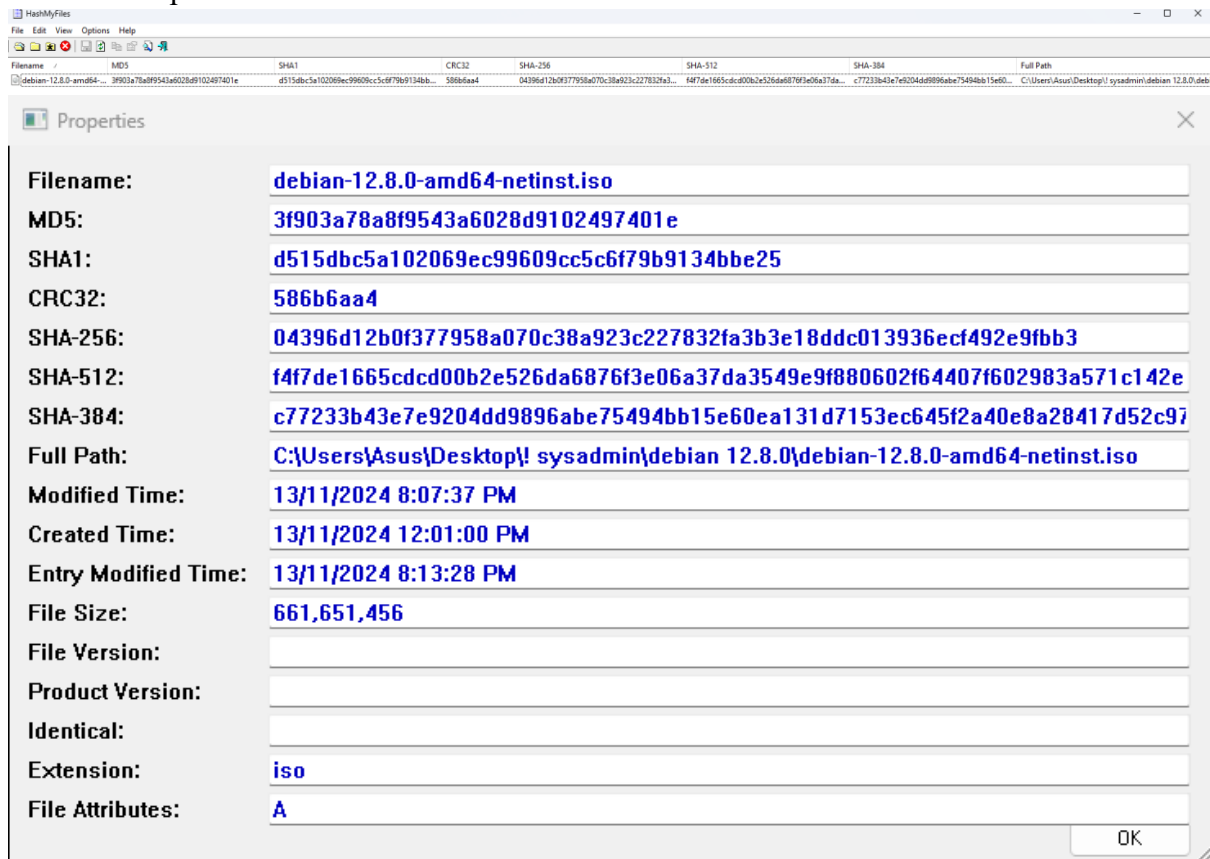
3. Most Linux distros are freely available and distributed in the format of ISO files. Normally it comes with MD5 or SHA sum file. Explain what ISO file is, MD5 and SHA sum.

ISO is identical copy or image of data found on an optical disc such as CD and DVD.

MD5 is message digest 5

SHA is secure hash algorithm or SHA sum.

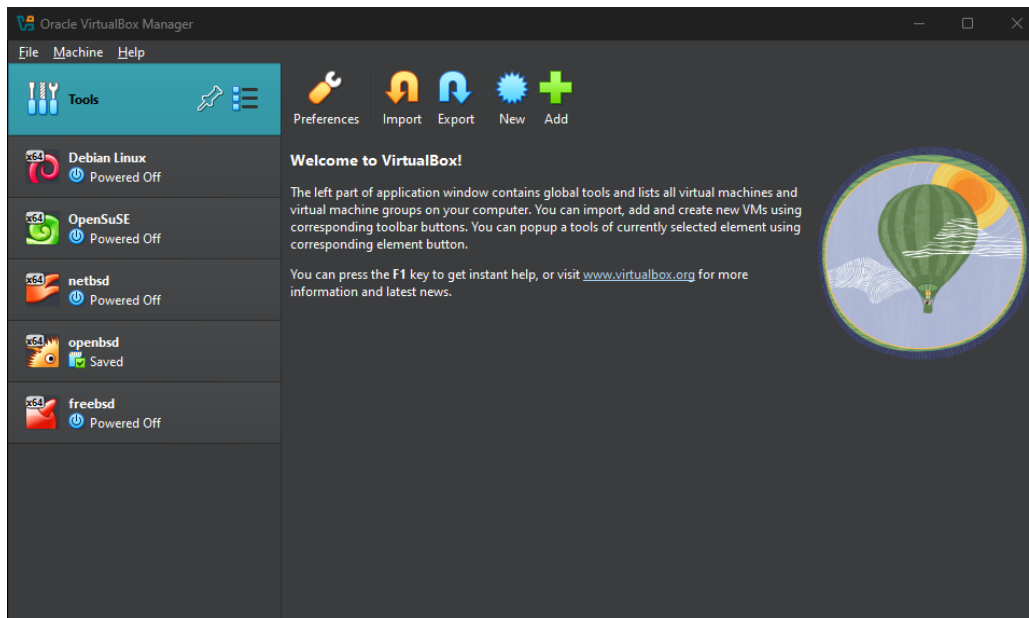
4. Download Debian ISO file and compute and verify their sum using certutil, HashMyFiles http://www.nirsoft.net/utils/hash_my_files.html, md5deep, mdsummer or any other program that can compute hash for a file.



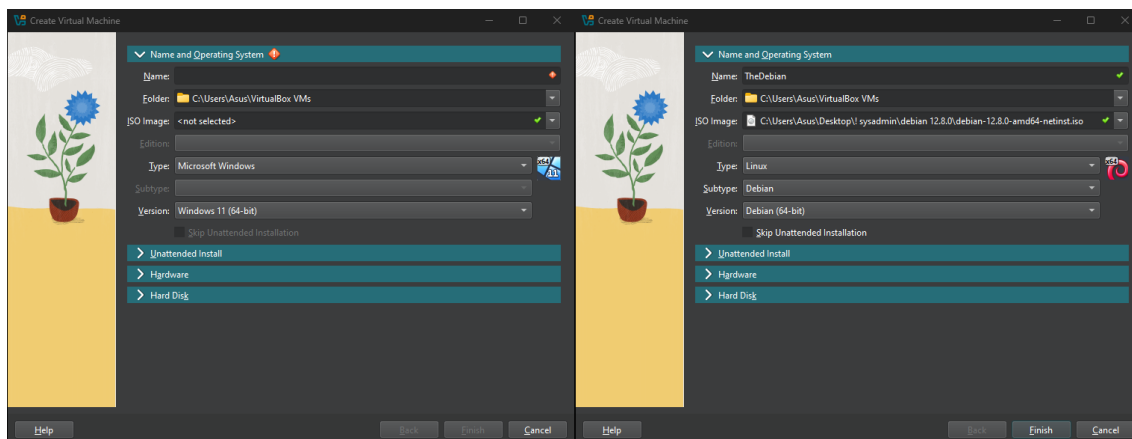
5. Install Debian Linux in your Lab PC using VirtualBox. In VM setup, set 2GB RAM and 10GB disk size. Capture the installation process.

Redo – need ss every stages and screen.

Click on New to create a new Debian Linux VM.



Under “Name and Operating System”, name the VM and select the Debian .iso file. Type, Subtype, and Version will automatically be selected after naming the VM.



Under “Unattended Install”, you can rename the username and password. In this example, username is “eusoff”, and password is “121112228”.

Create Virtual Machine

> Name and Operating System

✓ Unattended Install

Username and Password

Username: eusoff ✓

Password: 121112228 ✓

Repeat Password: 121112228 ✓

Additional Options

Product Key: #####-####-####-####-####

Hostname: TheDebian ✓

Domain Name: myguest.virtualbox.org ✓

☐ Install in Background

☐ Guest Additions

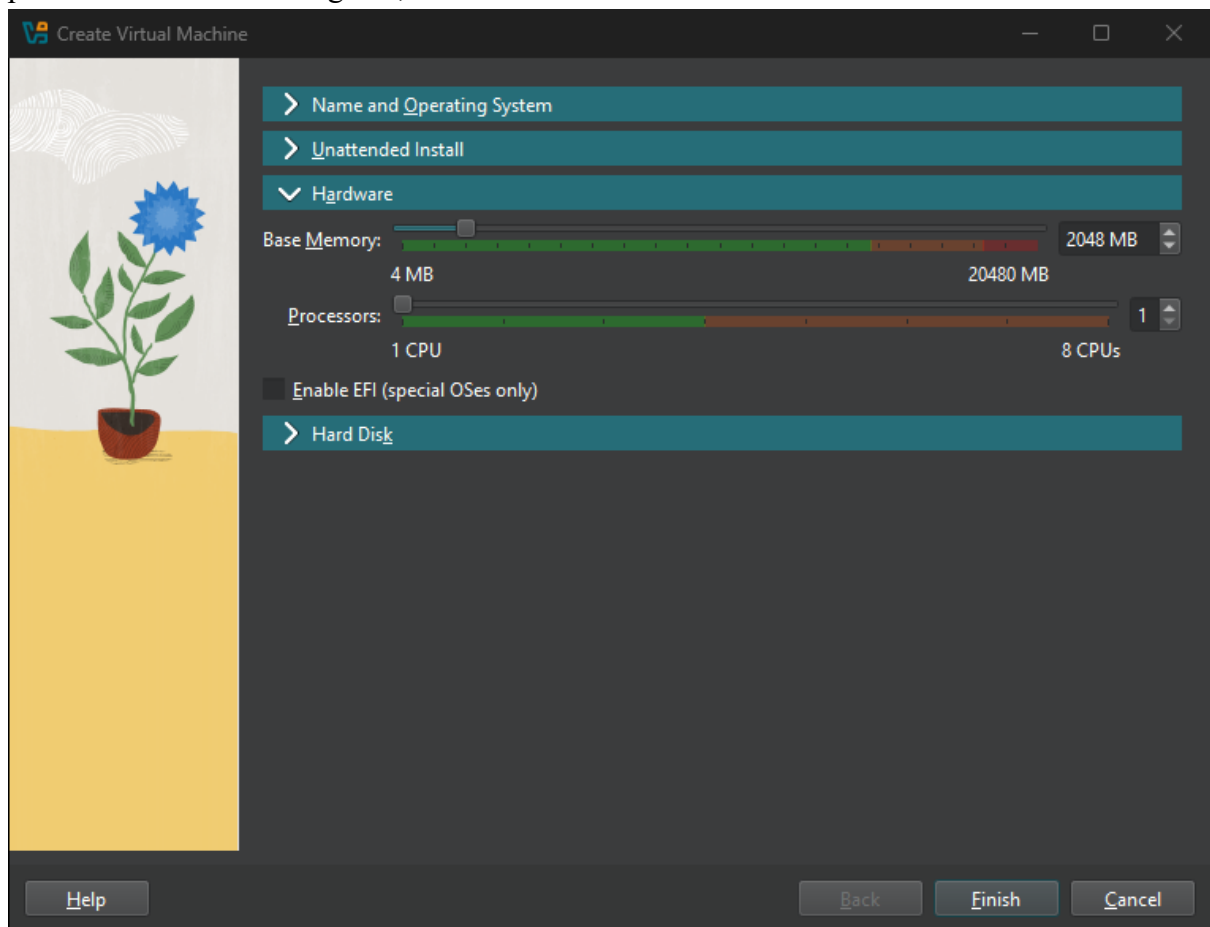
Guest Additions ISO: <not selected>

> Hardware

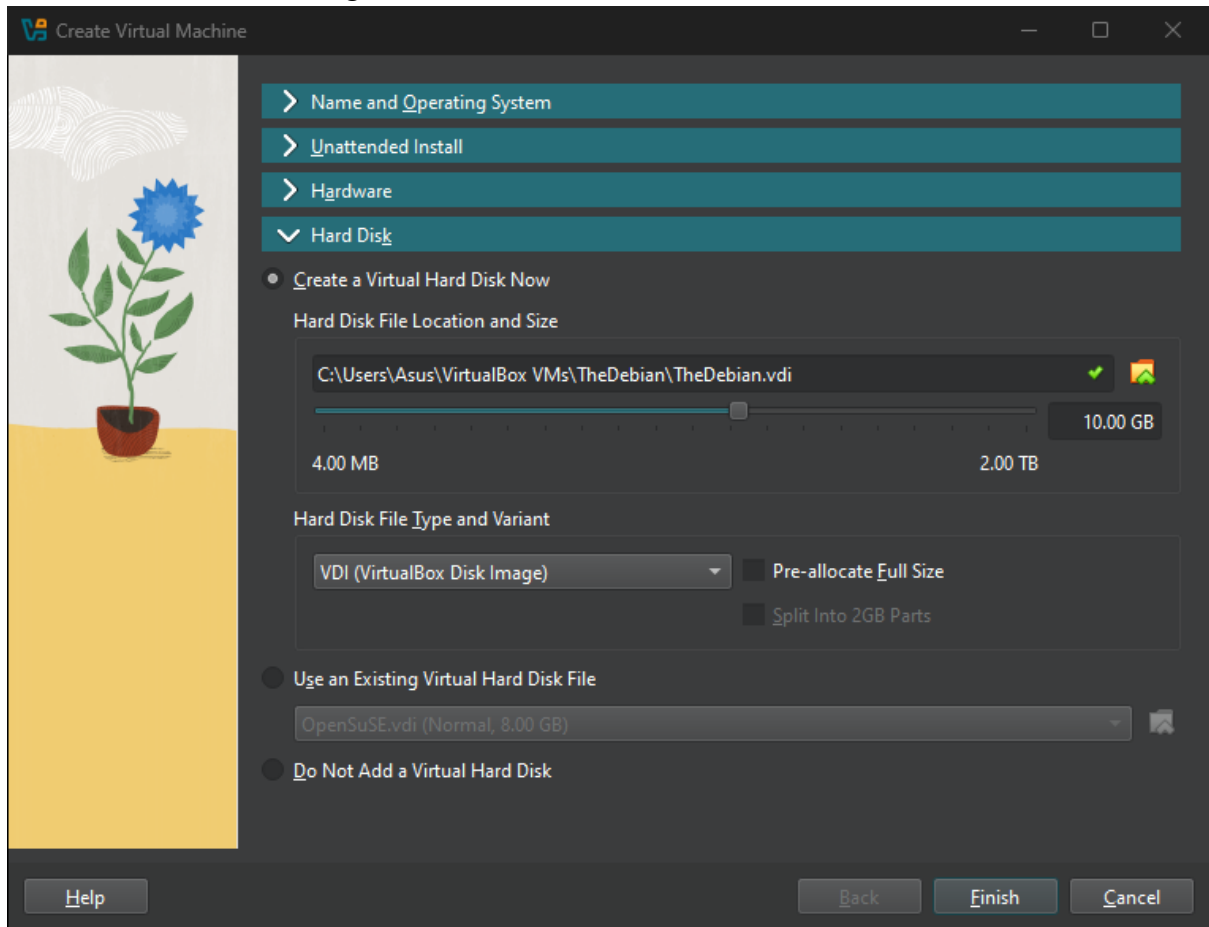
> Hard Disk

Help Back Finish Cancel

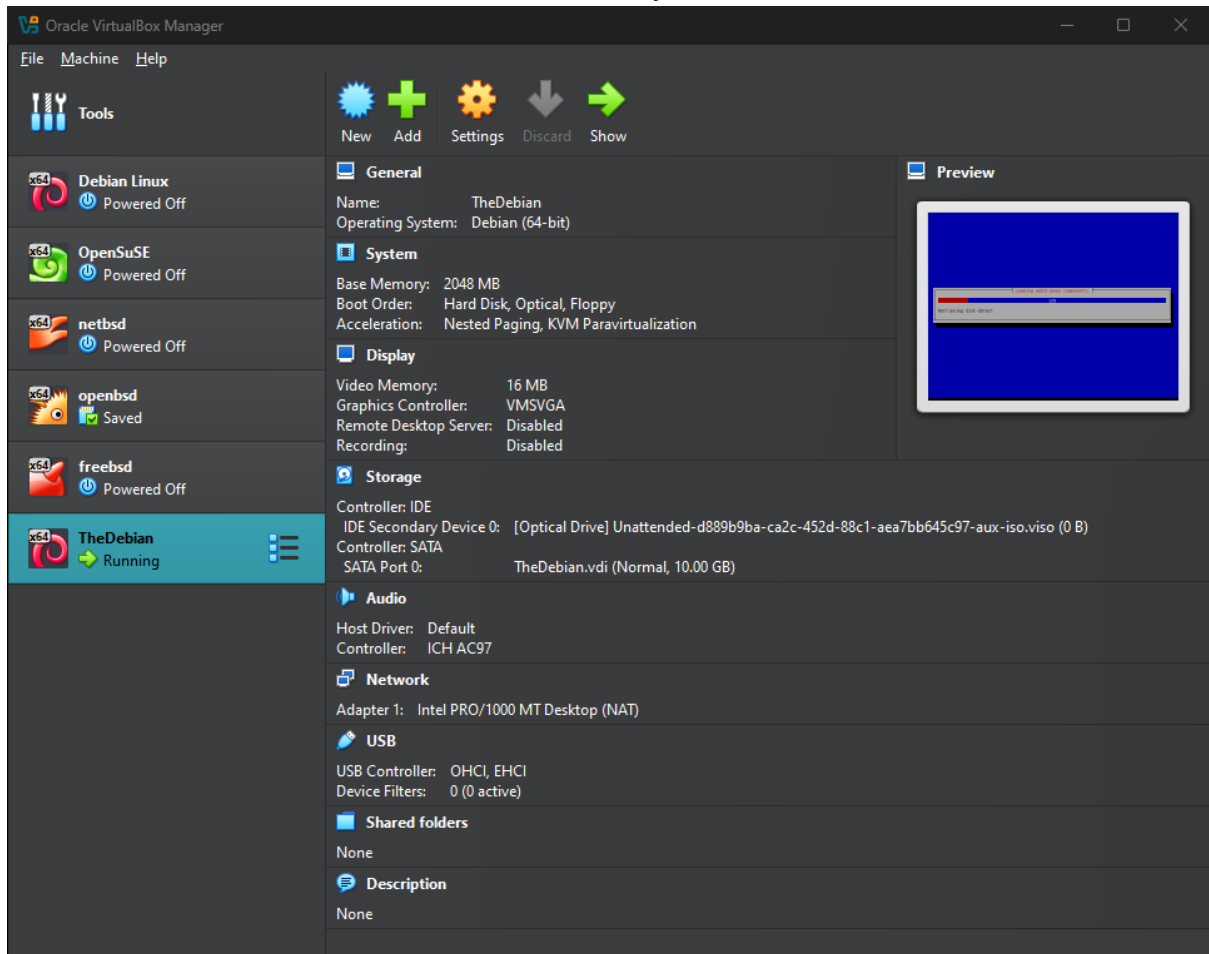
Under “Hardware”, we can set the base memory, which is RAM, and how much CPU processors count. In this guide, I will select 2GB RAM and 1 CPU.



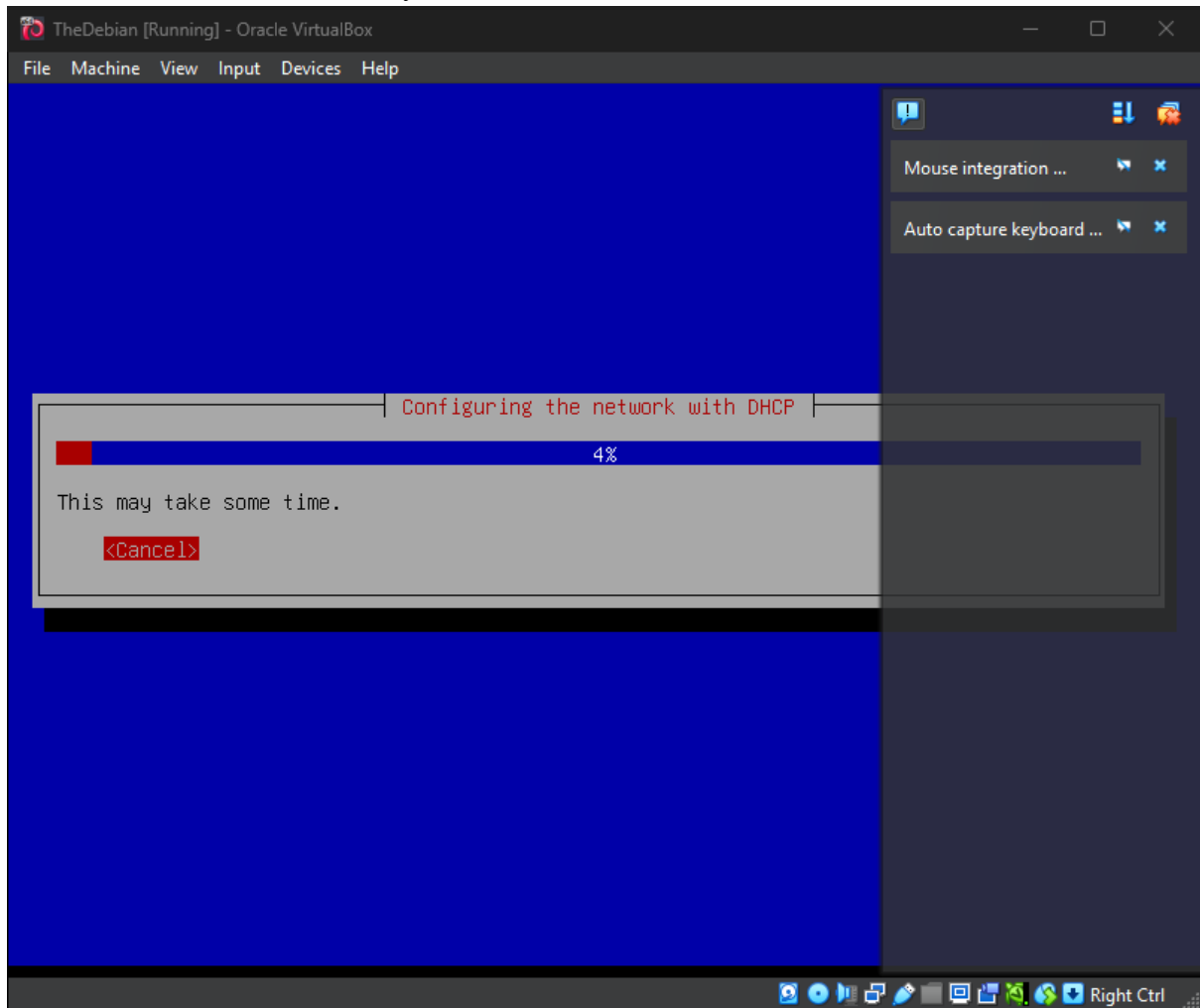
I will select 10GB of storage for this VM.



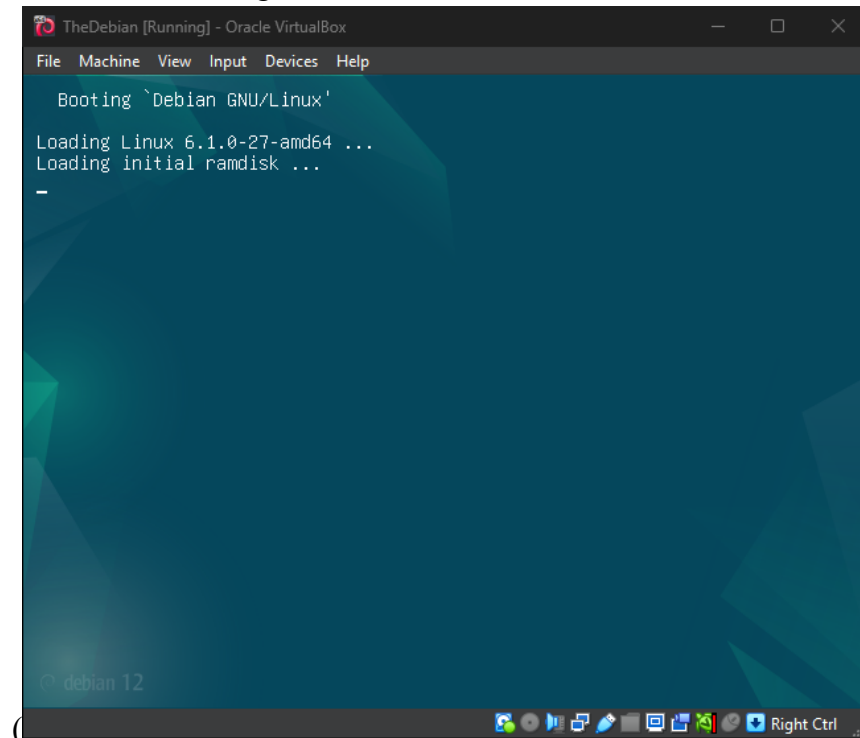
And then, click “Finish”. VM will ran automatically.



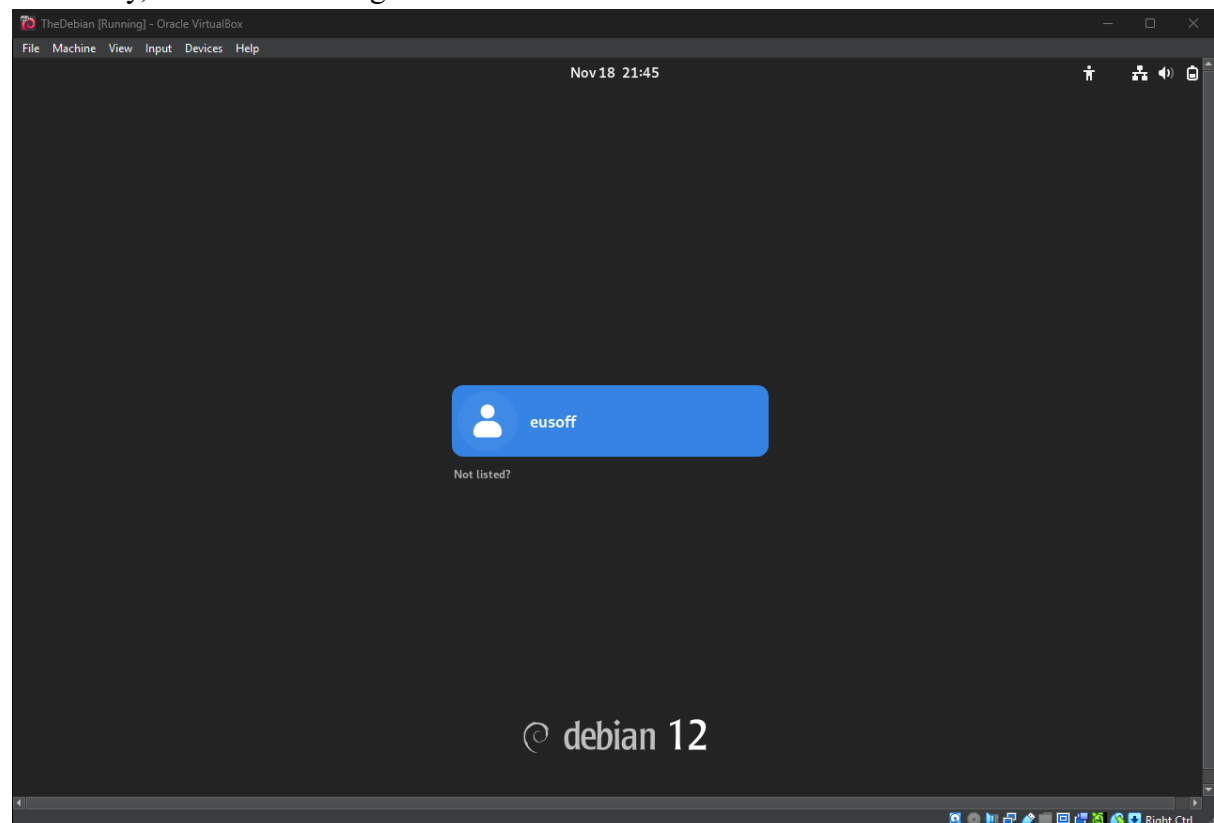
Let the VM run and install. May take some time.



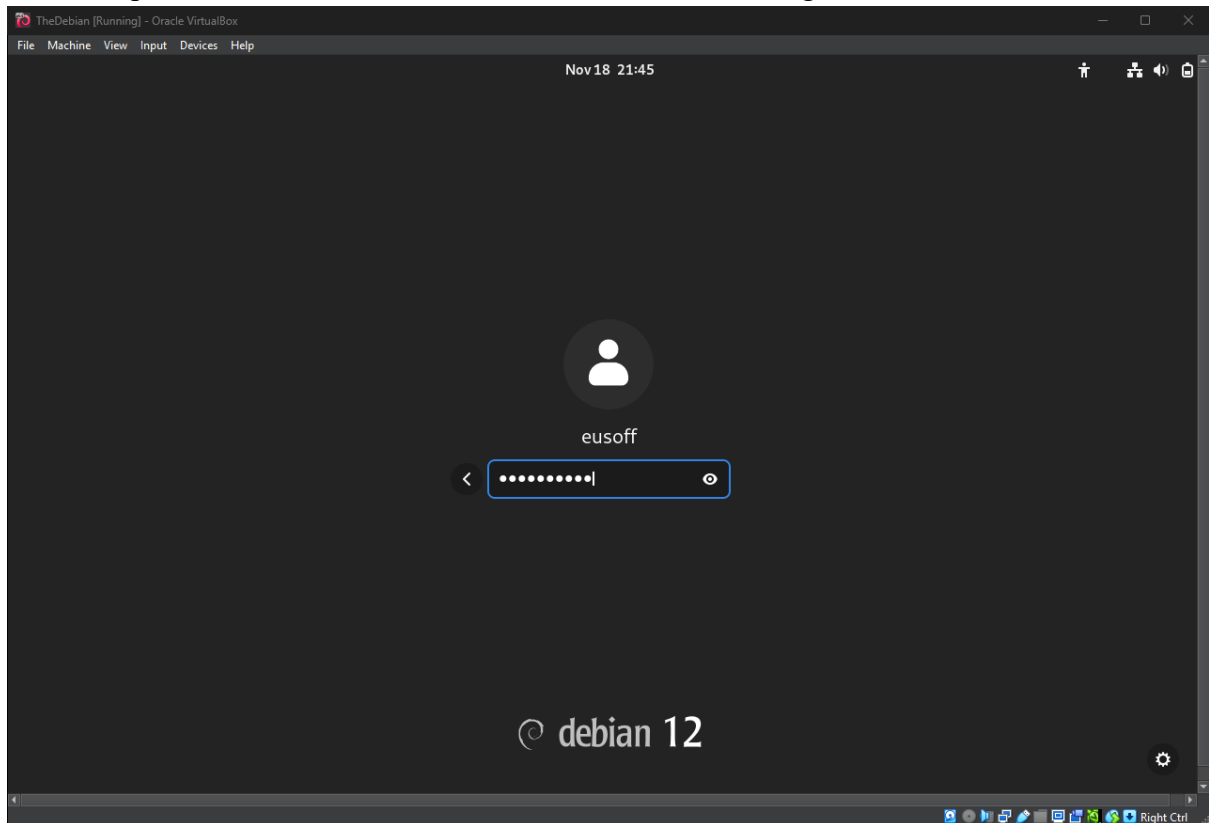
Once done installing, it will reboot and allow us to choose which bootloader to boot from.



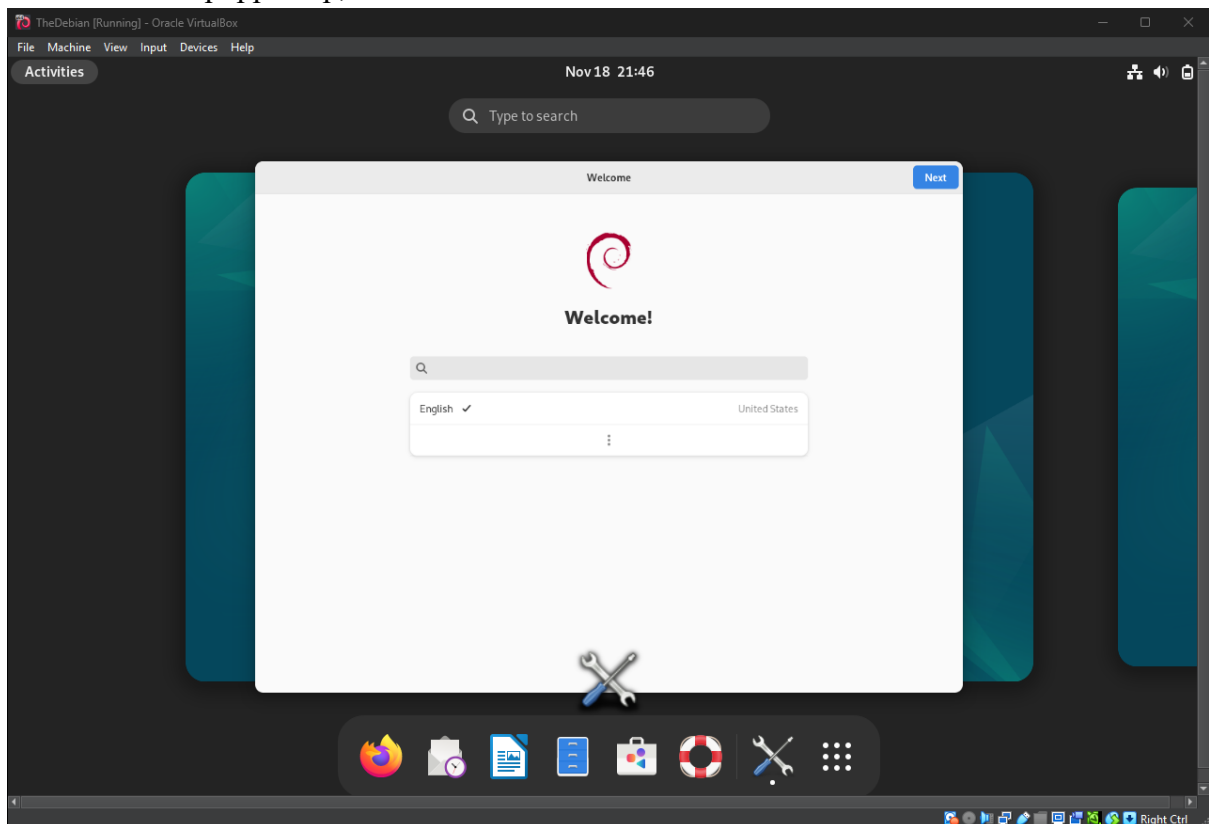
And lastly, we are able to login.



Enter the password for our username and we will be able to log in.



Once this have popped up, we are free to use Debian Linux.



Open setting, and click on About. It will show the specifications of the VM.

